AcadXpert: Acadmic manager

Minor Project – I Report

Submitted in partial fulfillment of the requirement for the award of the degree of

MASTER OF COMPUTER APPLICATION

SUBMITTED BY

NAME OF STUDENTS (14PT)

(UNIVERSITY ENROLMENT NO)

UNDER THE GUIDANCE OF

SUPERVISOR NAME (14PT)

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NAME

CERTIFICATE

This is to certify that this project entitled "ROZOB (JOB PORTAL)" submitted in partial fulfillment of the 1st semester of MCA to the VIPS, done by Mr./Ms. Siddharth Chandran and Harshal Kotey , Roll No. 35917704424 and 35117704424 is an authentic work carried out by him/her under my guidance.

SELF CERTIFICATE

This is to certify that the project report entitled “ROZOB (JOB PORTAL)” is done by me is an authentic work carried out for the partial fulfilment of the 1st semester of Master Computer Applications under the guidance of Dr. Dheeraj Malhotra. The matter embodied in this project work has not been submitted earlier for award of any degree or diploma to the best of my knowledge and belief.

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CHAPTER 1

Introduction of the project

Overview

Education plays a pivotal role in shaping society, and technology is increasingly becoming a key enabler in enhancing the efficiency of educational institutions. The "Academic Management System" is a comprehensive application designed to streamline and digitize the management of academic data such as attendance and student marks. This system caters to the needs of both faculty and students by providing them with a structured, interactive, and real-time platform for academic record management.

The system ensures efficient handling of semester-specific data, eliminates the need for manual record-keeping, and minimizes human errors in calculations and data organization. This project focuses on delivering a scalable, user-friendly, and efficient interface for managing core academic processes.

Problem Statement

Educational institutions often rely on manual processes or fragmented systems to handle academic records, which can lead to:

- Errors in attendance and marks entry.
- Increased workload for faculty due to repetitive tasks.
- Lack of transparency in data access for students.
- Inefficient tracking of semester-specific performance.

These challenges highlight the need for an integrated solution that facilitates seamless management of attendance, marks, and academic reports while ensuring real-time access to data.

Purpose of the Project

The primary goal of this project is to automate and optimize academic processes, enabling faculty and students to focus on their core objectives:

- For Faculty : Simplify tasks such as attendance management, marks recording, and report generation, reducing administrative overhead.

- For Students : Provide a transparent and accessible platform to monitor academic performance and attendance records.

This system fosters better communication between faculty and students while promoting accountability and efficiency.

Objectives

The project aims to:

1. Create separate login portals for students and faculty with role-specific functionalities.

2. Develop a real-time attendance management module that highlights students with attendance below a specified threshold (e.g., 75%).

3. Build a marks recording system that allows faculty to input, compile, and generate semester-specific reports.

4. Provide students with a user-friendly dashboard to access attendance and academic performance data.

5. Ensure the system operates locally, offering reliability and speed without dependency on external servers.

Scope of the Project

This project is designed to cater to the needs of small- to medium-sized educational institutions. Key functionalities include:

- Student Login :
  - View attendance records.
  - Check marks and reports for individual subjects.
- Faculty Login :
  - Take attendance and record marks.
  - Identify and flag students with attendance below the required percentage.
  - Compile and generate reports for individual students or entire classes.
- Semester-Specific Operations :
  - All data, including attendance and marks, is managed on a semester basis to maintain organized records.

In the future, this system can be scaled to include additional features like course management, notifications, and integration with external databases or cloud storage.

Importance and Benefits

The system offers several benefits to both faculty and students:

1. Automation : Reduces manual effort by automating repetitive tasks like calculating attendance percentages and compiling marks.

2. Accuracy : Minimizes errors in record-keeping and ensures accurate calculations.

3. Transparency : Enables students to view their academic performance and attendance records in real time.

4. Time-Saving : Speeds up administrative tasks, giving faculty more time to focus on teaching.

5. Scalability : Can be expanded to support more features or additional departments.

Features at a Glance

1. Separate Logins :

   - Faculty and students have distinct access with role-based permissions.

2. Attendance Management :

   - Faculty can take daily attendance.

   - Automatic identification of students with low attendance (below 75%).

3. Marks Management :

   - Faculty can input and compile marks for subjects.

   - Generate detailed reports for individual students or entire semesters.

4. Local Real-Time Operation :

   - Data is updated instantly and stored locally for speed and reliability.

5. Semester-Specific Functionality :

   - Attendance and marks are categorized semester-wise for organized data retrieval.

Relevance in Modern Education

With increasing demand for transparency and efficiency in education, this project aligns with the modern needs of digital transformation in academic institutions. It empowers faculty by automating tedious administrative tasks and enhances the student experience by providing timely access to academic data.

CHAPTER 2

MODULES IDENTIFIED

To implement the Academic Management System, the project is divided into several interdependent modules, each focusing on a specific functionality. Below are the modules identified, along with their roles and descriptions:

Authentication Module

- Purpose: Provides secure login functionality for students and faculty.
- Features:
  - Separate login portals for students and faculty.
  - Role-based authentication and authorization.
  - Password encryption for secure access.
  - Option to reset or recover passwords.

Student Management Module

- Purpose : Maintains a database of student information and semester-specific details.
- Features :
  - Add, update, or delete student details (e.g., name, ID, semester, and subjects).
  - Fetch student data for attendance and marks management.
  - View and manage semester-wise student records.

Faculty Management Module

- Purpose : Handles faculty-related information and roles.
- Features :
  - Add, update, or delete faculty information.
  - Assign faculty to specific subjects or classes.
  - Manage faculty roles and permissions within the system.

Attendance Management Module

- Purpose : Tracks and manages student attendance.
- Features :
  - Faculty can mark daily attendance for each class and subject.
  - View and edit attendance records for specific students or subjects.
  - Automatically calculate attendance percentages.
  - Highlight students with attendance below a specified threshold (e.g., 75%).
  - Export attendance reports if needed.

Marks Management Module

- Purpose : Records and compiles student marks for all subjects.
- Features :
  - Faculty can input marks for students on a subject-wise basis.
  - Store marks for assignments, midterms, and final exams.
  - Automatically calculate total marks and averages for each student.
  - Generate and export compiled marks reports.

Report Generation Module

- Purpose : Creates detailed academic reports for students and faculty.
- Features :
  - Generate semester-wise attendance and marks reports.
  - Export reports in formats like PDF or Excel for distribution or archival.
  - Provide customizable templates for report generation.

Semester Management Module

- Purpose : Organizes data based on semester-specific requirements.
- Features :
  - Categorize attendance and marks by semester.
  - Provide separate dashboards for different semesters.
  - Ensure data is archived and retrieved semester-wise.

File Upload and Data Import Module

- Purpose : Allows bulk upload of attendance or marks via Excel files.
- Features :
  - Provide a standardized template for Excel uploads.
  - Validate data before importing to ensure accuracy.
  - Handle errors and display skipped rows or issues during the upload process.

Real-Time Data Update Module

- Purpose : Ensures that all updates are reflected immediately across the system.
- Features :
  - Automatically refresh data on dashboards upon changes.
  - Notify users of updates in real time.

Dashboard Module

- Purpose : Serves as the primary interface for users.
- Features :
  - Provide faculty with an overview of classes, attendance, and marks.
  - Display students' attendance percentage and academic performance.
  - Summarize key semester-specific details in an intuitive layout.

Database Management Module

- Purpose : Handles all data storage and retrieval operations.
- Features :
  - Maintain structured databases for students, faculty, attendance, and marks.
  - Enable efficient querying and updates to ensure system responsiveness.
  - Support data backup and restoration features.

CHAPTER 3

Flow graph of project

CHAPTER 4

Tools/Technologies for Implementation

The implementation of the Academic Management System involves a full-stack development approach, leveraging modern technologies for front-end, back-end, database management, and deployment. Below is a categorized list of tools and technologies suitable for the project:

Front-End Technologies

These technologies will be used to create an interactive and user-friendly interface for students and faculty.

- HTML5 : For structuring web pages and defining the content layout.
- CSS3 : For styling the application, ensuring it is visually appealing and responsive.
- JavaScript : For implementing client-side logic and enhancing interactivity.
- Frameworks/Libraries :
  - React.js : To build dynamic and reusable UI components.
  - Bootstrap : To ensure a responsive and mobile-friendly design.

Back-End Technologies

The back-end will handle application logic, authentication, and communication with the database.

- Programming Language :
  - Node.js : A JavaScript runtime for developing server-side logic.
  - Java with Spring Boot (if preferred): For robust backend API development.
- Frameworks :
  - Express.js : Lightweight framework for building APIs with Node.js.
  - Spring Boot : For building enterprise-grade REST APIs (if Java is chosen).
- Authentication :
  - JSON Web Tokens (JWT) : For secure session management and authentication.
  - OAuth2 : For advanced authentication mechanisms.

Database Technologies

A robust database is essential for storing and managing student, faculty, attendance, and marks data.

- Relational Database :
  - MySQL or PostgreSQL : For structured, reliable, and relational data storage.
- NoSQL Database (optional):
  - MongoDB : For handling unstructured or semi-structured data, if needed.
- ORM (Object-Relational Mapping) :
  - Mongoose (for MongoDB) or JPA/Hibernate (for SQL databases).

File Management and Data Import

To handle file uploads and data imports, these tools and libraries will be utilized:

- Libraries :
  - Multer (for Node.js): For managing file uploads like Excel sheets.
  - Apache POI (for Java): For reading and writing Excel files.
- File Formats Supported :
  - Excel (XLS/XLSX) : For bulk uploads of attendance and marks data.
  - CSV : For lightweight data imports and exports.

Deployment and Hosting

The system will be deployed locally or on a cloud platform for accessibility.

- Local Deployment :
  - XAMPP/WAMP : For local server hosting during development.
- Cloud Deployment (optional):
  - Heroku , AWS , or Azure : For cloud-based deployment.
  - Docker : For containerized deployment.
- Version Control :
  - Git : For managing source code.
  - GitHub/GitLab : For remote repository hosting and collaboration.

CHAPTER 5

References